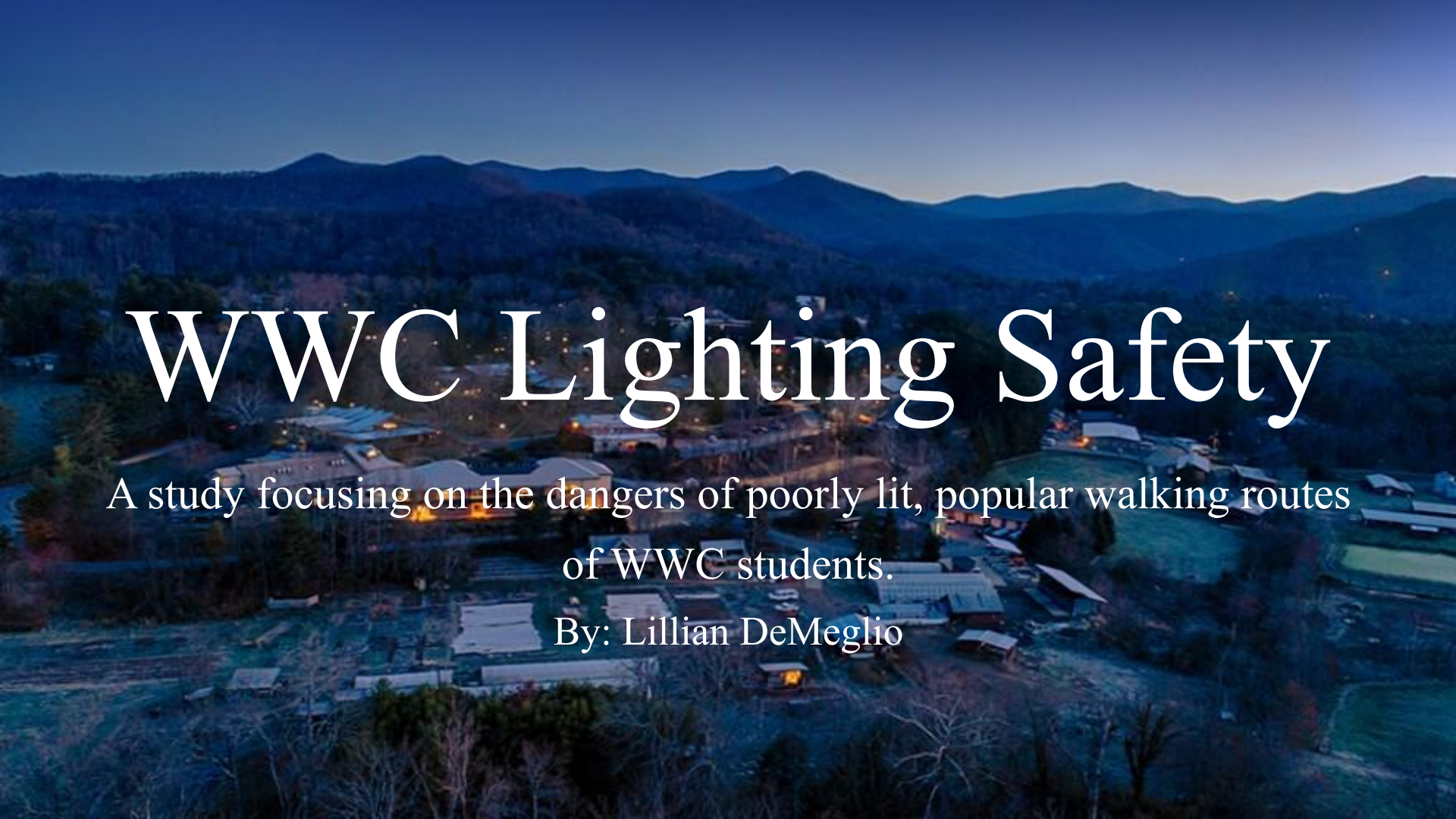


WWC Lighting Safety

A study focusing on the dangers of poorly lit, popular walking routes of WWC students.

By: Lillian DeMeglio

Background

This study aims to target the issue with Warren Wilson's campus lighting. My maps were created in hopes of bringing awareness to the lack of well-lit paths that students travel the most from the available parking lots to their dorms, as well as suggest possible solutions to this problem.

Data

In order to determine what is safe and unsafe lighting, I researched data on Lumen production in lights. To understand Lumens and light production, here are some important definitions:

Lumen: Lumens measure brightness, higher lumens indicate a brighter light.

Foot Candle: The amount of illumination produced by a candle at a distance of one foot away. One ft-c equals 1 lumen/square foot.

OSHA Standards and Information

- ★ According to OSHA, the minimum amount of light that a bulb must produce in order for it to be considered safe is 100-200 lumens. This is specific to outdoor lighting, and after doing some calculations, I derived that the minimum (100 in this case) lumens will have a circumference of 141.37 feet.
- ★ In order to conduct this study, I had to travel each path on the upper side of campus, from Ballfields to Kitteridge, collecting the coordinates of each post I encountered and recording them on my phone.

Warren Wilson Lighting



Methods Used

- ★ Coordinate collection using Google Maps and Excel Spreadsheets
- ★ Shapefiles including WWC buildings and boundary
- ★ Conversion from Delimited Text to Geopackage
- ★ Buffer tool to create the circumference of projected light
- ★ Fixed Geometries to create the most common walking routes of students, highlight student dorms, and highlight the parking lots available for student use

Analysis and Results pt.1

Safest Route for Students:

Parking right outside of Devries gym taking the route to Schaeffer dorms along the paved road *NOT* the shortcut beside Shepard dorm.

Students impacted:

- ★ Upper classmen have the advantage of a well-lit route to their housing.
- ★ Underclassmen residing in the designated freshmen dorms will have a much longer path to take regardless of which lot they park in.

Analysis and Results pt.2

Riskiest Route for Students:

- ★ Parking in Dust Bowl and traveling along the soccer fields to arrive at Ballfields/Eco dorms.
- ★ Parking in Kitteridge lot and traveling to Villages; while a shorter route, is extremely unlit.

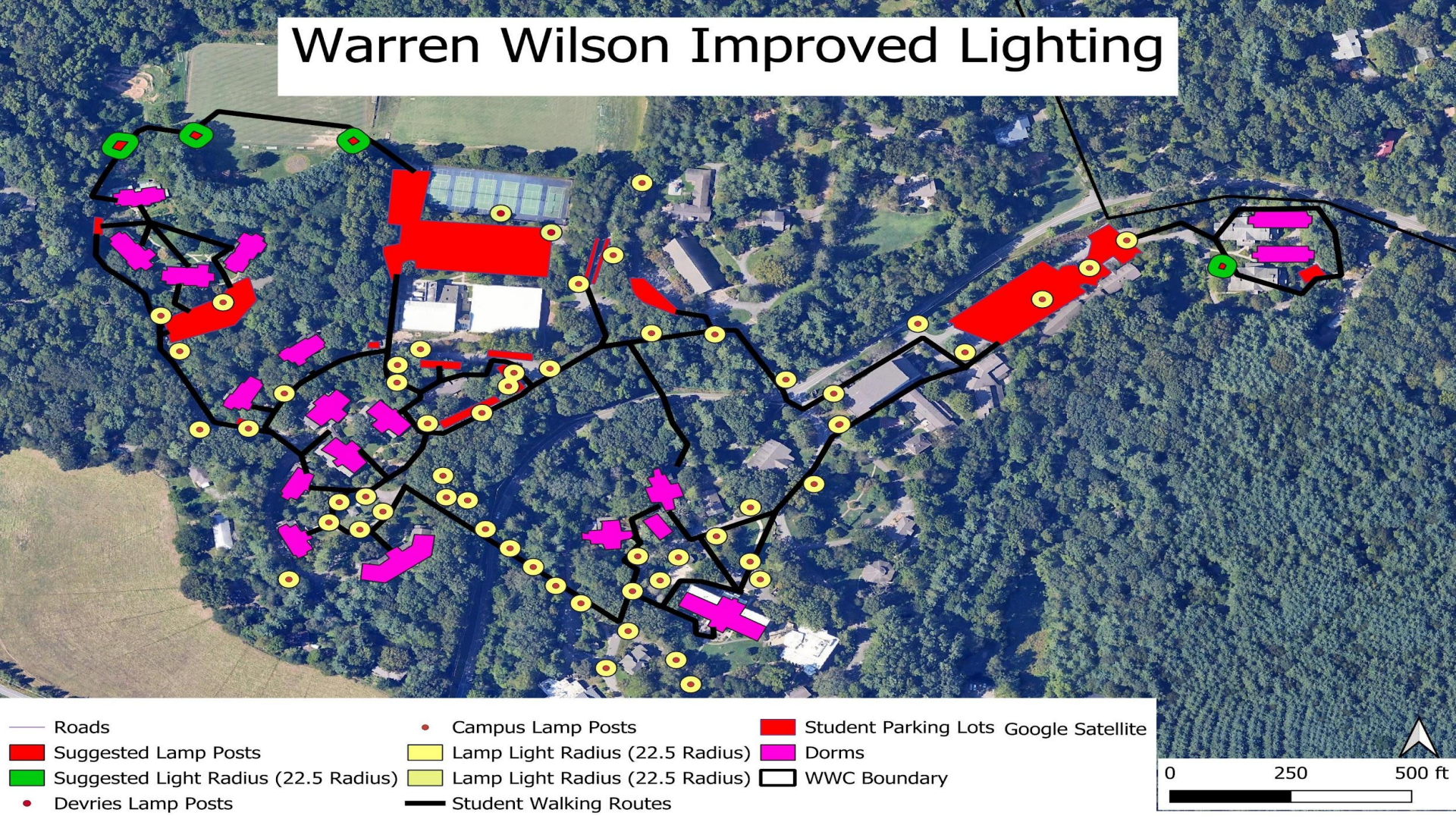
Students impacted:

- ★ Upper classmen residing in Ballfields or Villages are more suseptible to wildlife and potentially harmful people because of the thick forest that surrounds both routes and dormitories.

Suggestions and Improvements

- ★ I recommend adding at least four new lamp posts on campus, which are illustrated in the map on the next slide.
- ★ Routine lighting checks to update old, dim, or completely inoperable lights which were found in my study.
- ★ Because of the variety of different light posts and their heights, and more inclusive Lumen study on the campus lighting should be done.

Warren Wilson Improved Lighting




Thank you!
And please stay safe on your campus journey!

Sources for Lighting Information

- -https://www.tcpi.com/osha-lighting-standards-for-workplace-safety/
- -https://www.softlights.org/lighting-recommendations/#:~:text=The%202700K%2C%20820%20lumen%20light,2600%20lumen%20is%20truly%20dangerous.